

Research Notes on Principal component analysis

Principal component analysis

>> Section 1

Principal component analysis creates variables that are linear combinations of the original variables. The new variables have the property that the variables are all orthogonal. The PCA transformation can be helpful as a pre-processing step before clustering. PCA is a variance-focused approach seeking to reproduce the total variable variance, in which components reflect both common and unique variance of the variable. PCA is generally preferred for purposes of data reduction (that is, translating variable space into optimal factor space) but not when the goal is to detect the latent construct or factors. Factor analysis is similar to principal component analysis, in that factor analysis also involves linear combinations of variables. Different from PCA, factor analysis is a correlation-focused approach seeking to reproduce the inter-correlations among variables, in which the factors "represent the common variance of variables, excluding unique variance". In terms of the correlation matrix, this corresponds with focusing on explaining the off-diagonal terms (that is, shared co-variance), while PCA focuses on explaining the terms that sit on the diagonal. However, as a side result, when trying to reproduce the on-diagonal terms, PCA also tends to fit relatively well the off-diagonal correlations. Results given by PCA and factor analysis are very similar in most situations, but this is not always the case, and there are some problems where the results are significantly different. Factor analysis is generally used when the research purpose is detecting data structure (that is, latent constructs or factors) or causal modeling. If the factor model is incorrectly formulated or the assumptions are not met, then factor analysis will give erroneous results.

>> Section 2

$$w(1) = \arg \max_{\mathbf{w}} \mathbf{w} = 1 \{ \mathbf{w}^T \mathbf{X} \mathbf{w} = 2 \} = \arg \max_{\mathbf{w}} \mathbf{w} = 1 \{ \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} \}$$

>> Section 3

K-means clustering It has been asserted that the relaxed solution of k-means clustering, specified by the cluster indicators, is given by the principal components, and the PCA subspace spanned by the principal directions is identical to the cluster centroid subspace. However, that PCA is a useful relaxation of k-means clustering was not a new result, and it is straightforward to uncover counterexamples to the statement that the cluster centroid

subspace is spanned by the principal directions.

>> Section 4

where $\Sigma^{\wedge}$ is the square diagonal matrix with the singular values of X and the excess zeros chopped off that satisfies $\Sigma^{\wedge 2} = \Sigma^T \Sigma$. Comparison with the eigenvector factorization of XTX establishes that the right singular vectors W of X are equivalent to the eigenvectors of XTX , while the singular values $\sigma(k)$ of X are equal to the square-root of the eigenvalues $\lambda(k)$ of XTX . Using the singular value decomposition the score matrix T can be written

>> Section 5

Quantitative finance In quantitative finance, PCA is used in financial risk management, and has been applied to other problems such as portfolio optimization. PCA is commonly used in problems involving fixed income securities and portfolios, and interest rate derivatives. Valuations here depend on the entire yield curve, comprising numerous highly correlated instruments, and PCA is used to define a set of components or factors that explain rate movements, thereby facilitating the modelling. One common risk management application is to calculating value at risk, VaR, applying PCA to the Monte Carlo simulation.

>> Section 6

where $\mathbf{y}$ is a q -element vector and $\mathbf{B}'$ is a $(q \times p)$ matrix, and let $\Sigma_{\mathbf{y}} = \mathbf{B}' \Sigma \mathbf{B}$ be the variance-covariance matrix for $\mathbf{y}$. Then the trace of $\Sigma_{\mathbf{y}}$, denoted $\text{tr}(\Sigma_{\mathbf{y}})$, is maximized by taking $\mathbf{B} = \mathbf{A}_q$, where $\mathbf{A}_q$ consists of the first q columns of $\mathbf{A}$ ($\mathbf{B}'$ is the transpose of $\mathbf{B}$). ($\mathbf{A}$ is not defined here) Property 2: Consider again the orthonormal transformation

>> Section 7

Directional component analysis Directional component analysis (DCA) is a method used in the atmospheric sciences for analysing multivariate datasets. Like PCA, it allows for dimension reduction, improved visualization and improved interpretability of large data-sets. Also like PCA, it is based on a covariance matrix derived from the input dataset. The difference between PCA and DCA is that DCA additionally requires the input of a vector direction,

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referred to as the impact. Whereas PCA maximises explained variance, DCA maximises probability density given impact. The motivation for DCA is to find components of a multivariate dataset that are both likely (measured using probability density) and important (measured using the impact). DCA has been used to find the most likely and most serious heat-wave patterns in weather prediction ensembles, and the most likely and most impactful changes in rainfall due to climate change.

>> Section 8

Matrix V , also of dimension $p \times p$, contains p column vectors, each of length p , which represent the p eigenvectors of the covariance matrix C . The eigenvalues and eigenvectors are ordered and paired. The j th eigenvalue corresponds to the j th eigenvector. Matrix V denotes the matrix of right eigenvectors (as opposed to left eigenvectors). In general, the matrix of right eigenvectors need not be the (conjugate) transpose of the matrix of left eigenvectors.

>> Section 9

so each column of T is given by one of the left singular vectors of X multiplied by the corresponding singular value. This form is also the polar decomposition of T . Efficient algorithms exist to calculate the SVD of X without having to form the matrix XTX , so computing the SVD is now the standard way to calculate a principal components analysis from a data matrix, unless only a handful of components are required. As with the eigen-decomposition, a truncated $n \times L$ score matrix TL can be obtained by considering only the first L largest singular values and their singular vectors: